WORKED SOLUTION

STEP II Specimen paper, Question 1

Pure • Partial Fractions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q1

The key idea

Before reaching for the usual partial-fractions machinery, look at the numerator: it is exactly the derivative of the denominator. Since the denominator factorises as a product of linear factors, f is a logarithmic derivative, and the logarithmic derivative of a product is the sum of the logarithmic derivatives of the factors. For the second part, the equation becomes a sum of three unit fractions equal to 1, and size alone forces x to be tiny.

Solution

The partial fractions

The denominator factorises:

$$D(x) = x^3 + (a+b)x^2 + abx = x(x^2 + (a+b)x + ab) = x(x+a)(x+b),$$

and differentiating gives

$$D'(x) = 3x^2 + 2(a+b)x + ab,$$

which is precisely the numerator. So $\mathrm{f}(x) = D'(x)/D(x)$.

General case, $a \neq b$. The three factors x, $x+a$, $x+b$ are distinct (they differ because $a \neq 0$, $b \neq 0$ and $a \neq b$), so we expect three simple partial fractions. Using the product rule,

$$D'(x) = (x+a)(x+b) + x(x+b) + x(x+a),$$

and dividing by $D(x) = x(x+a)(x+b)$ term by term gives

$$\boxed{\mathrm{f}(x) = \frac{1}{x} + \frac{1}{x+a} + \frac{1}{x+b}} \qquad (a \neq b).$$

(Equivalently, $\mathrm{f}(x) = \dfrac{\mathrm{d}}{\mathrm{d}x}\ln|x(x+a)(x+b)|$, and the logarithm of a product splits into a sum.) Of course the cover-up rule gives the same thing: at $x = 0$ the residue is $ab/(ab) =$

1, and at $x = -a$ it is $(3a^2 - 2a^2 - 2ab + ab)/((-a)(b-a)) = a(a-b)/(a(a-b)) = 1$, and similarly at $x = -b$.

Ask yourself What goes wrong with this form when $a = b$?

Special case, $a = b$. Now the denominator is $x(x+a)^2$, with a repeated factor, so the standard form would be $\frac{A}{x} + \frac{B}{x+a} + \frac{C}{(x+a)^2}$. But the numerator also has a factor $x + a$:

$$3x^2 + 4ax + a^2 = (3x + a)(x + a),$$

so it cancels, leaving

$$\mathrm{f}(x) = \frac{3x+a}{x(x+a)} = \boxed{\frac{1}{x} + \frac{2}{x+a}} \qquad (a = b),$$

where the last step is the cover-up rule ($x = 0$ gives $a/a = 1$; $x = -a$ gives $-2a/(-a) = 2$). There is no $(x+a)^{-2}$ term at all, so $C = 0$. This agrees with the general formula with b replaced by a, but it is worth noticing that the repeated factor has disappeared rather than producing a squared term.

Since a and b are non-zero, the factor x can never coincide with $x + a$ or $x + b$, so these are the only cases.

When $\mathrm{f}(x) = 1$

Now let x, a, b be positive integers. In both cases above (the $a = b$ formula is just the general one with $b = a$),

$$\mathrm{f}(x) = \frac{1}{x} + \frac{1}{x+a} + \frac{1}{x+b},$$

and all three terms are positive. Because $a, b \geq 1$, the last two terms are each strictly less than $\frac{1}{x}$.

Ask yourself How large can x be if three fractions, each at most $\frac{1}{x}$, add up to 1?

- If $x = 1$, then $\mathrm{f}(x) = 1 + (\text{positive}) > 1$.
- If $x \geq 4$, then $\mathrm{f}(x) < \frac{3}{x} \leq \frac{3}{4} < 1$.
- If $x = 3$, then $x + a \geq 4$ and $x + b \geq 4$, so $\mathrm{f}(x) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{4} = \frac{5}{6} < 1$.

So $\mathrm{f}(x) = 1$ forces $x = 2$, and then we need

$$\frac{1}{2+a} + \frac{1}{2+b} = \frac{1}{2}.$$

Multiplying by $2(2+a)(2+b)$ gives $2(2+b) + 2(2+a) = (2+a)(2+b)$, that is $8 + 2a + 2b = 4 + 2a + 2b + ab$, so

$$ab = 4.$$

With a, b positive integers the only possibilities are $(a,b) = (1,4), (2,2), (4,1)$. Hence $f(x) = 1$ exactly for

$$(x,a,b) = (2,1,4), \quad (2,2,2), \quad (2,4,1),$$

which is a finite number (three) of values, as required. Note that $(2,2,2)$ is the special case $a = b$: $\frac{1}{2} + \frac{2}{4} = 1$.

Checks

The three solutions are the familiar unit-fraction identities $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$ and $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$. Numerically, the partial-fraction forms were compared with the original f in exact rational arithmetic at 2000 random rational points (for both $a \neq b$ and $a = b$), and a brute-force search over all $1 \leq x, a, b < 200$ found exactly the three triples above and no others.

Common slips

- Ploughing into the $a = b$ case with a $\frac{C}{(x+a)^2}$ term and not noticing the cancellation; you should find $C = 0$, and spotting the factor $(x + a)$ in the numerator saves the work.
- Forgetting to rule out $x = 1$ and $x = 3$; the crude bound $f < 3/x$ only disposes of $x \geq 4$.
- Stopping at "$ab = 4$" without listing the solutions, or listing only $(1,4)$ and forgetting $(4,1)$ and $(2,2)$.

Beyond STEP

The identity $D'/D = \sum 1/(x - r_i)$, where the r_i are the roots of a polynomial D (repeated roots counted with multiplicity), is a workhorse: it underlies Newton's identities for power sums of roots and, in complex analysis, the argument principle, which counts the zeros of a function inside a curve by integrating f'/f around it. Equations like $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$ in positive integers are Egyptian-fraction problems, a classic corner of Diophantine equations where bounding arguments like ours do all the work. Needham's *Visual Complex Analysis* gives a beautiful geometric account of the argument principle, and Davenport's *The Higher Arithmetic* is a good place to start on Diophantine equations.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.